17 (2023) 100275

Contents lists available at ScienceDirect

International Journal of Thermofluids

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Entropy analysis of the MHD Jeffrey fluid flow in an inclined porous pipe with convective boundaries

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ARTICLE INFO

Keywords:
Entropy generation
Jeffrey fluid
Convective boundary conditions
Suction/injection
Porous pipe
Magnetohydrodynamics

ABSTRACT

The present work deals with entropy generation analysis of a non-Newtonian Jeffrey fluid flow through an inclined pipe of circular cross section in the presence of uniform porous medium. The flow is subjected to constant pressure gradient and uniform magnetic field. The boundaries of circular pipe are assumed to be convective. Due to the presence of mixed convection, the underlying governing equations are nonlinear and coupled. The numerical solution of the system of governing equations is obtained. The analysis of roles played by the transport properties viz. fluid velocity and temperature is done to explore the thermodynamic entities like entropy generation number and Bejan number. The study of irreversibilities in the considered flow is analysed taking several sets of parameters, i.e., the Jeffrey fluid parameter, the Hartmann number, the Darcy number, the Grashof number, suction/injection parameter, the Brinkman number, the Prandtl number and the Biot number appearing in the problem.

1. Introduction

The Newtonian fluid model was inadequate to describe some physiological fluids like human blood, lipids, saliva etc. Studying these fluids has a crucial role in understanding the human body's several processes. Along with the biological fluid flows, some fluids are used in many industries that do not behave as Newtonian fluids. In this regard, several models of fluids were introduced with a nonlinear relationship between stress and rate of strain, and these fluids are called non-Newtonian fluids [1–3]. Out of many models of non-Newtonian fluids, one promising model is the model of Jeffrey fluid. The model of Jeffrey fluid is an important generalization of the Newtonian fluid model in which the time derivatives are taken in place of convective derivatives. The above model explains the concept of relaxation and retardation time efficiently. The aforementioned physiological fluids can be described aptly using Jeffrey fluid theory [4,5]. Several authors worked in the advancement of Jeffrey fluid flows. In the articles [6,7], authors studied Jeffrey fluid flow in narrow tubes, in the presence of magnetic field and through a porous medium, respectively. Devakar et al. [8] considered the flow of Jeffrey fluid in a straight uniform square duct. Turkyilmazoglu and Pop [9] obtained the theoretical solution in a setup of a parallel external flow with stretching/shrinking sheet for the flow and heat transfer of a Jeffrey fluid near the stagnation point. In a stretching sheet setup, Zeeshan and Majid [10] included suction/injection to study the heat transfer aspects of Jeffrey fluid. Recently, Aleem et al. [11] considered the generalized boundary conditions for MHD flow of Jeffrey fluid with heat transfer.

Over the past several decades, researchers worldwide have been working on the challenging problem of discovering the ways of optimum use of energy in the process of transport of fluid and heat. The second law of thermodynamics plays an important role in this process, as it states that all real processes are irreversible. As can be referred from [12–14], the entropy generation, due to various sources, majorly concerns the thermodynamic irreversibility. The factors such as heat transfer across finite temperature gradients, characteristic of convective heat transfer and viscous dissipation play a key role in the phenomenon of irreversibility. The study of entropy generation has been proven effective in turbo-machinery, electric cooling and combustion, modern electronic systems, solar power collectors etc. Researchers have worked in several contexts to study entropy generation for efficient utilization of the energy in various fluid flows [15–21]. The analysis of entropy generation in the flows through pipes has a significant role in the engineering of heating and cooling devices, fluid distribution pipelines and crude oil transport. In this regard, the following literature can be found. Al-Zaharnah [22] studied the entropy analysis in pipe flow subjected to external heating. The authors in the study [23] investigated the entropy generation due to laminar forced convection in the entrance region

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https://doi.org/10.1016/j.ijft.2022.100275
Received 29 October 2022; Received in revised form 19 December 2022; Accepted 24 December 2022
Available online 27 December 2022
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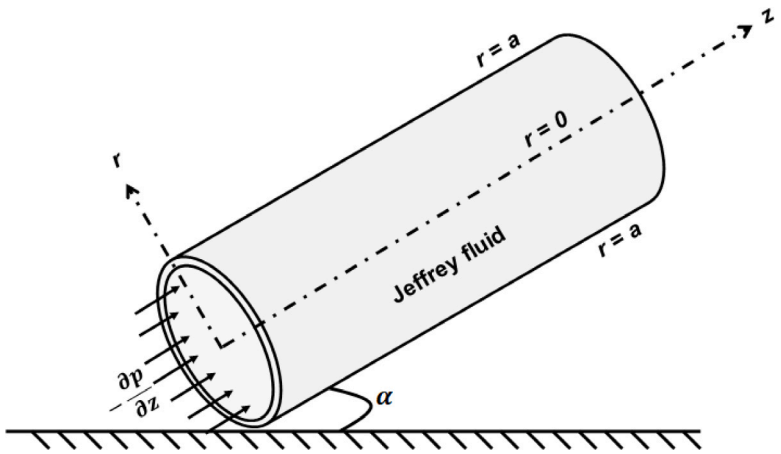


Fig. 1. Physical sketch of the problem.

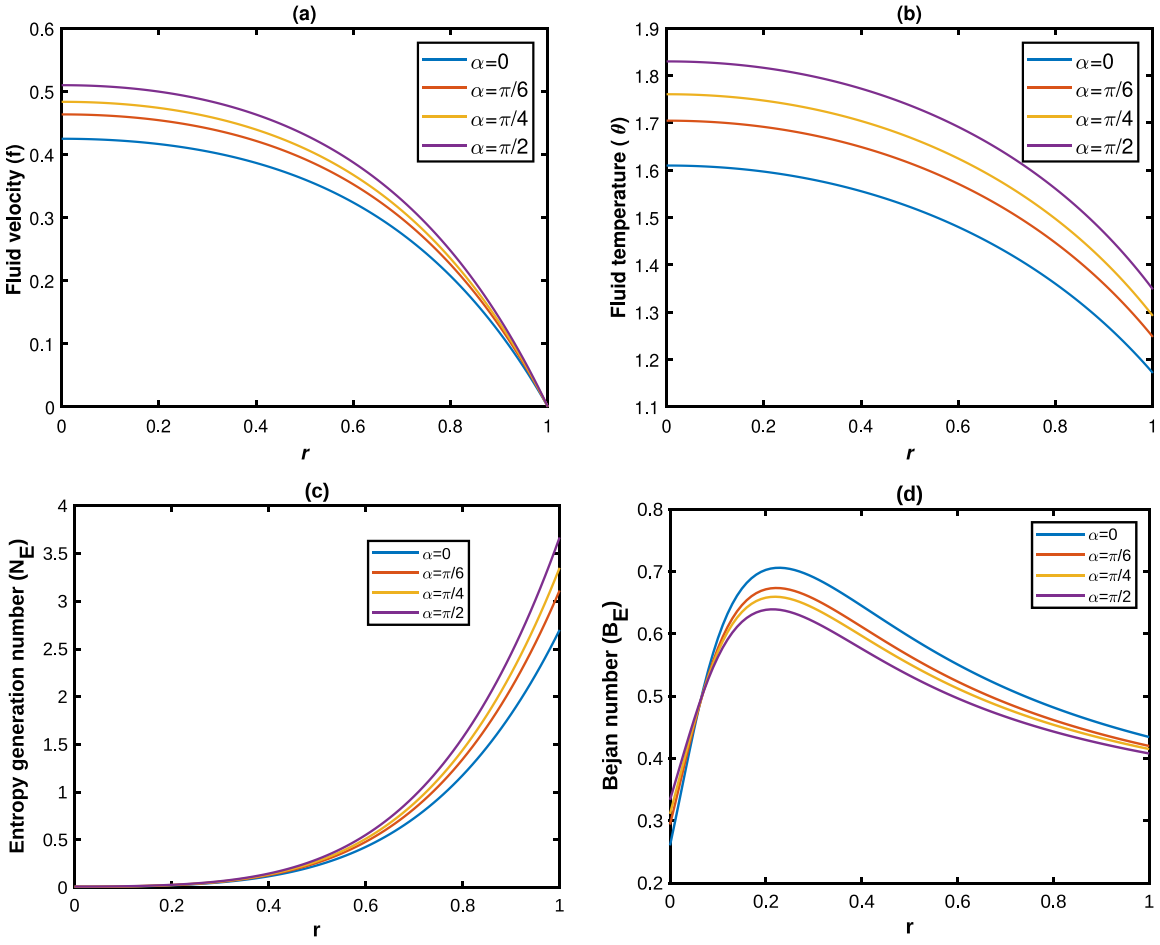


Fig. 2. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of inclination angle α.

of a concentric annulus. Ben-Mansour and Sahin [24] discussed the entropy generation in developing laminar fluid flow through a circular pipe. The entropy generation in ice slurry pipe flow is investigated by Bouzid et al. [25]. Nagaraju et al. [26] analysed the second law for the flow in a circular pipe with uniform suction and magnetic field effects. Nagaraju et al. [27] studied the entropy generation analysis of the MHD flow of couple stress fluid between two concentric rotating cylinders with porous lining. Srinivasacharya and Hima Bindu [28] presented the study of entropy generation of micropolar fluid flow in an inclined porous pipe with convective boundary conditions. Habibishandiz and Saghir [29] considered a setup of a vertical annular porous cylinder for the MHD mixed convection heat transfer of nanofluid with oxytactic microorganisms.

The applications like geophysics, earthquake engineering, heating/cooling of electronic devices etc., motivated the researchers to study the flows of electrically conducting fluids in the presence of magnetic effects. Especially when the flow is through circular pipe, the problem becomes more and more relevant. The pioneering work of Gold [30] reported the initial findings regarding MHD flows through pipes. Later on, many studies [31–39] have been carried out by various

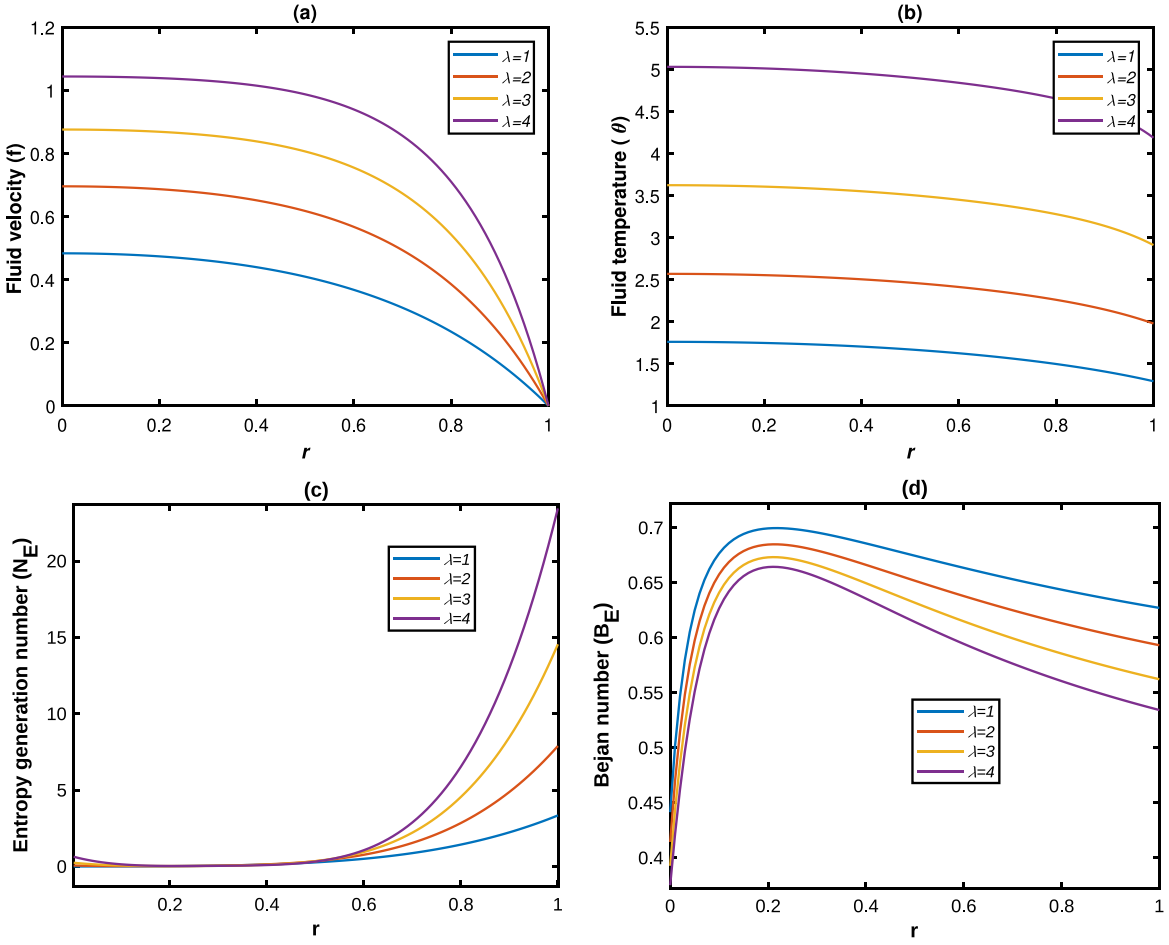


Fig. 3. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of Jeffrey fluid parameter λ.

researchers for MHD flow of fluids through pipes. The models consisting of porous pipes for fluid flow with suction/injection have been studied for past several years as these flows are pertinent in several engineering and biological processes, especially in designing bio-medical applications. Several studies [40–46] have emphasized the fluid flows through porous medium in various set ups.

In addition to the aforementioned studies for entropy generation for various flow geometries and constraints, the literature has also been reported for entropy generation specific to Jeffrey fluid flows [47–49]. However, none have addressed the entropy generation analysis for the MHD flow of Jeffrey fluid through inclined porous pipe employing convective boundary conditions. Addressing this un-attempted flow problem is the objective of the present work. The novelty of this work is that it contains the modelling, solution, and analysis of the flow and heat transfer of Jeffrey fluid along with the associated entropy generation analysis. The major parameters of interest are fluid velocity, fluid temperature, entropy generation and Bejan number. Considering the cylindrical geometry for an inclined circular pipe, the convective boundary conditions, magnetohydrodynamic effects, presence of porous medium together to study the entropy generation for the Jeffrey fluid is the new contribution to academic society as per our knowledge. The organization of the article is as follows: Section 1 deals with introduction of the flow problem along with the literature review. Section 2 contains the governing equations for the flow of Jeffrey fluid and heat transfer followed by the mathematical formulation of the problem along with boundary conditions, and subsequent non-dimensionalization. The entropy generation analysis has been carried out in Section 3, and the solution of the problem is obtained in Section 4. Section 5 gives the insight of obtained results and their explanation. Finally, in last section, the significant conclusions have been presented.

2. Governing equations and mathematical formulation

Consider the laminar, axi-symmetric and steady flow of Jeffrey fluid through the inclined pipe of radius a, and α is the inclination angle between circular pipe and surface (see Fig. 1). The magnetic field of strength H_0 is applied in the transverse direction to the pipe, and the fluid is assumed to be conducting when subjected to magnetic field. The density, viscosity, thermal conductivity, specific heat capacity of the Jeffrey fluid under consideration is ρ, μ, K, C_p, respectively, and p is the fluid pressure. The circular pipe under consideration is assumed to be filled with uniform porous medium of permeability k^*. The model has the setup of convective heat transfer using a hot fluid which induces a heat transfer coefficient h, assuming the temperature of the hot fluid to be T_1 and the surrounding temperature to be T_0. The cylindrical polar coordinates (r, θ, z) are considered to describe the dynamics of flow and heat transfer. The governing equations for two-dimensional flow of Jeffrey fluid are given by [6],

Equation of continuity

$$\frac{\partial u}{\partial r} + \frac{u}{r} + \frac{\partial u}{\partial z} = 0, \tag{1}$$

Equations of motion

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + w\frac{\partial u}{\partial z}\right) = -\frac{\partial p}{\partial r} + \frac{1}{r}\frac{\partial}{\partial r}(rS_{rr}) + \frac{\partial}{\partial z}(S_{rz}) + \rho g\beta(T - T_0)\cos\alpha - \sigma {H_0}^2 u - \frac{\mu}{k^*}u, \tag{2}$$

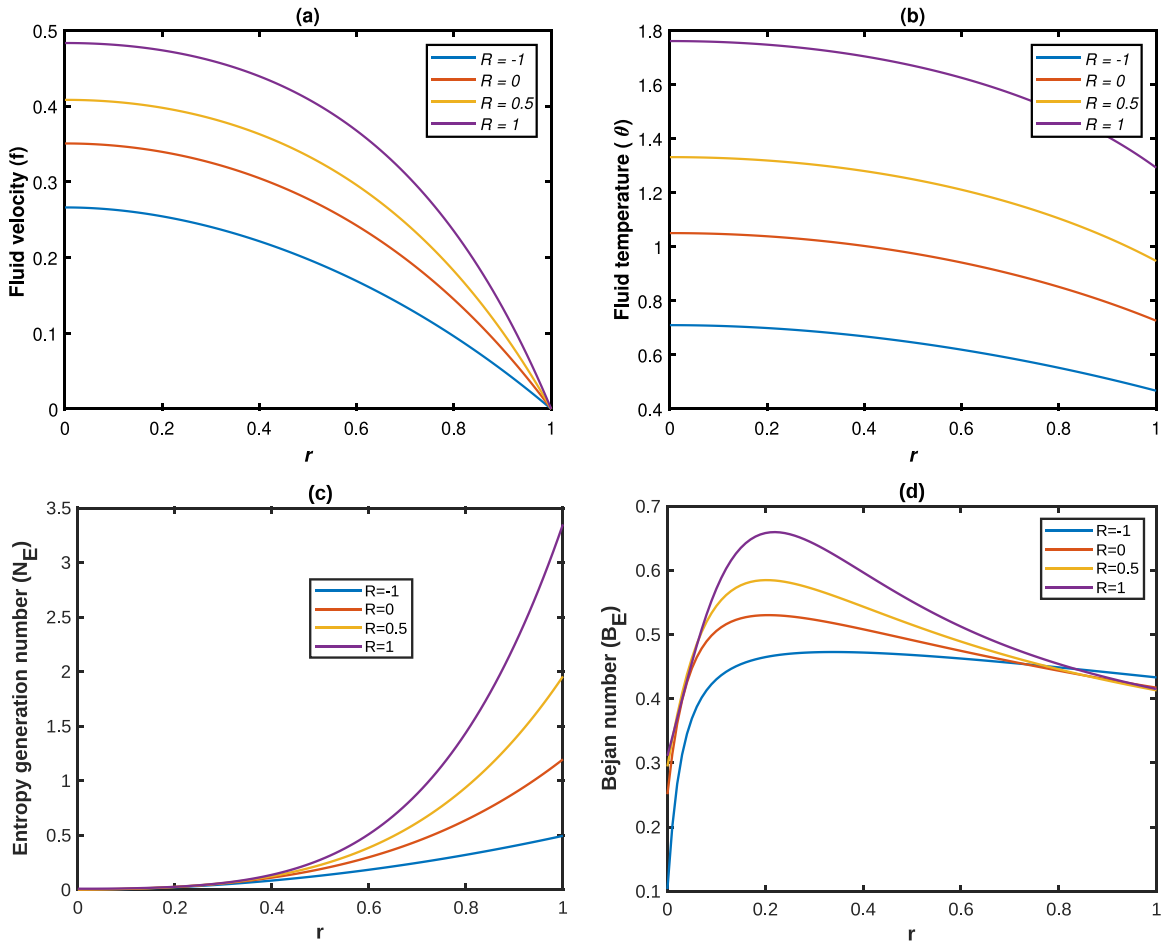


Fig. 4. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of suction/injection parameter R.

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial r}+w\frac{\partial w}{\partial z}\right)=-\frac{\partial p}{\partial z}+\frac{1}{r}\frac{\partial}{\partial r}(rS_{zr})+\frac{\partial}{\partial z}(S_{zz})+\rho g\beta(T-T_0)\sin\alpha-\sigma {H_0}^2 w-\frac{\mu}{k^*}w, \tag{3}$$

Equation of energy

$$\rho C_p\left(\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial r}+w\frac{\partial T}{\partial z}\right)=K\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)+\frac{\partial^2 T}{\partial z^2}\right]+\mu\phi, \tag{4}$$

where

$$\phi=\frac{4}{3}\left[\left(\frac{\partial u}{\partial r}\right)^2+\frac{u^2}{r^2}-\frac{u}{r}\frac{\partial w}{\partial r}\right]+\left(\frac{\partial w}{\partial r}\right)^2. \tag{5}$$

Here, S_{rr}, $S_{rz}(S_{zr})$ and S_{zz} are stress components, given in simplified form as,

$$\begin{aligned}S_{rr}&=\frac{2\mu}{1+\lambda_1}\left[\frac{\partial u}{\partial r}+\lambda_2 u\frac{\partial^2 u}{\partial r^2}+\lambda_2 w\frac{\partial^2 u}{\partial r\partial z}\right],\\ S_{rz}=S_{zr}&=\frac{\mu}{1+\lambda_1}\left[\frac{\partial w}{\partial r}+\frac{\partial u}{\partial z}+\lambda_2 u\frac{\partial^2 w}{\partial r^2}+\lambda_2 u\frac{\partial^2 u}{\partial r\partial z}+\lambda_2 w\frac{\partial^2 w}{\partial r\partial z}+\lambda_2 w\frac{\partial^2 u}{\partial z^2}\right],\\ S_{zz}&=\frac{2\mu}{1+\lambda_1}\left[\frac{\partial w}{\partial z}+\lambda_2 u\frac{\partial^2 w}{\partial r\partial z}+\lambda_2 w\frac{\partial^2 w}{\partial z^2}\right].\end{aligned} \tag{6}$$

The pipe is assumed to be of infinite length and flow is fully developed and steady. That is, the flow is considered in z-direction only and the fluid velocity is represented as $\overline{q}=(0,0,w(r))$. This velocity automatically satisfies incompressibility condition (1). Also, the transpiration velocity u_0 is constant, where $u_0>0$ is the injection velocity and $u_0<0$ is the suction velocity. The temperature is assumed as sole function of r i.e. $T=T(r)$. Imposing the same on the set of Eqs. (1)–(5), and making use of the stress components, we get the reduced set of differential equations as,

Equation of motion

$$\rho u_0\frac{dw}{dr}=-\frac{dp}{dz}+\frac{\mu}{1+\lambda_1}\left[\frac{1}{r}\frac{d}{dr}\left(r\frac{dw}{dr}\right)\right]-\sigma {H_0}^2 w-\frac{\mu}{k^*}w+\rho g\beta(T-T_0)\sin\alpha, \tag{7}$$

Equation of energy

$$K\left[\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right)\right]+\mu\left(\frac{dw}{dr}\right)^2=\rho C_p u_0\frac{dT}{dr}, \tag{8}$$

The no-slip boundary condition at the boundary of pipe is assumed along with the convective boundary conditions. Along the axis of the pipe($r=0$) the regularity conditions for fluid velocity and temperature is taken. Mathematically, these conditions are expressed as,

$$\begin{aligned}u&=0 \quad \text{at} \quad r=a,\\ -K\frac{dT}{dr}&=h(T-T_0) \quad \text{at} \quad r=a,\\ \frac{dw}{dr}&=0 \quad \text{at} \quad r=0,\\ \frac{dT}{dr}&=0 \quad \text{at} \quad r=0,\end{aligned} \tag{9}$$

where h is the heat transfer coefficient.

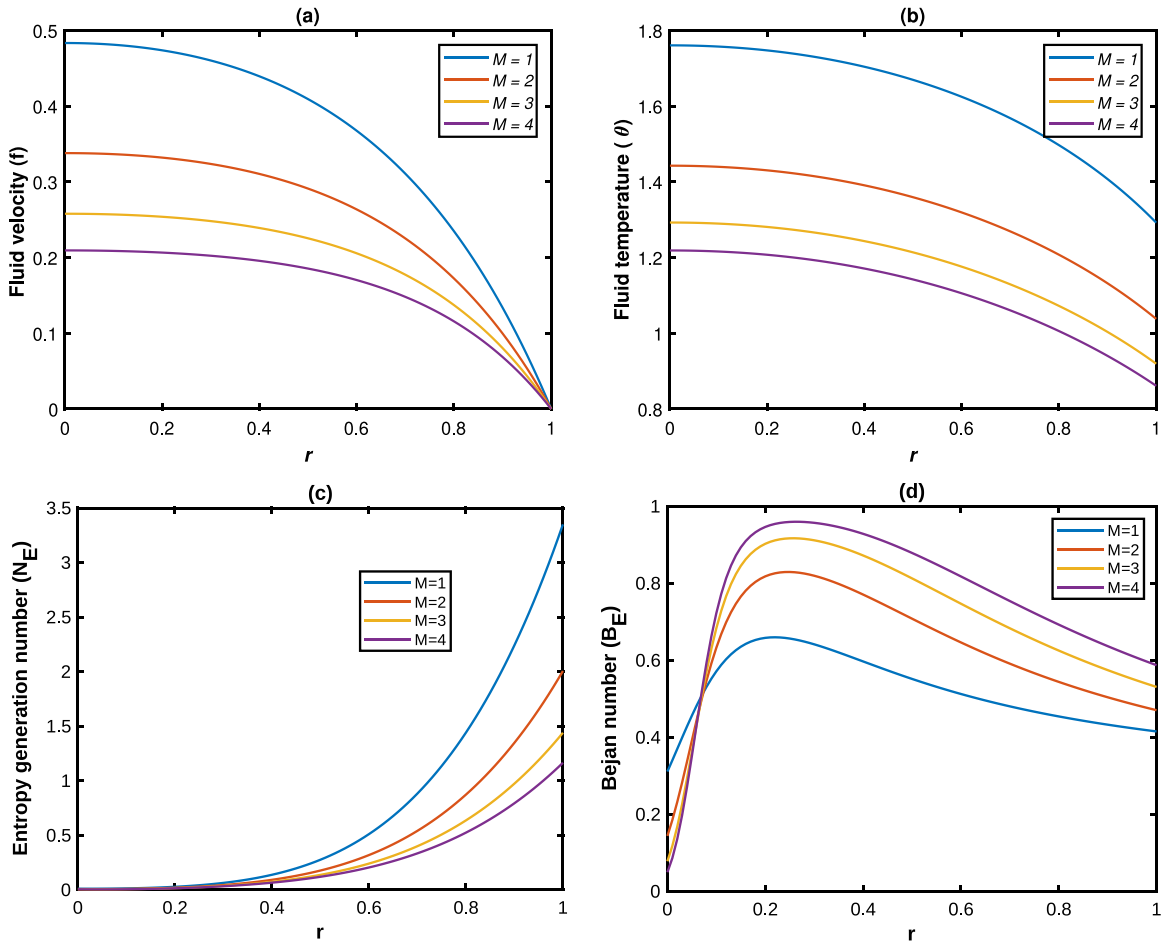


Fig. 5. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of Hartmann number M.

Introducing the following variables to non-dimensionalize the system of governing equations [19],
$r = a\eta$, $w = w_0 f(\eta)$, $\theta(\eta) = \dfrac{T - T_0}{T_1 - T_0}$, we get the governing equations,

$$G + \frac{1}{1+\lambda_1}\left(f'' + \frac{1}{\eta}f'\right) - \left(M + \frac{1}{Da}\right)f + \frac{Gr}{Re}\sin(\alpha)\theta = Rf', \tag{10}$$

$$\theta'' + \frac{1}{\eta}\theta' + B_R(f')^2 = RP_R\theta', \tag{11}$$

along with the corresponding conditions,

$$\begin{aligned} f(\eta) &= 0 \quad \text{at} \quad \eta = 1, \\ \theta'(\eta) + Bi\theta(\eta) &= 0 \quad \text{at} \quad \eta = 1, \\ f'(\eta) &= 0 \quad \text{at} \quad \eta = 0, \\ \theta'(\eta) &= 0 \quad \text{at} \quad \eta = 0, \end{aligned} \tag{12}$$

where, G is the constant pressure gradient, λ_1 is the Jeffrey fluid parameter for ratio of relaxation to retardation times, $M = \sqrt{\dfrac{\sigma {H_0}^2 a^2}{\mu}}$ is the Hartmann number, $Da = \dfrac{k^*}{a^2}$ is the Darcy number, $G_R = \dfrac{\rho^2 g\beta(T_1 - T_0)a^3}{\mu^2}$ is the Grashof number, $Re = \dfrac{\rho a w_0}{\mu}$ is the Reynolds number, $R = \dfrac{\rho a u_0}{\mu}$ is the suction/injection parameter, $B_R = \dfrac{\mu {w_0}^2}{K\Delta T}$ is the Brinkman number, $P_R = \dfrac{C_p\mu}{K}$ is the Prandtl number and $Bi = \dfrac{ah}{K}$ is the Biot number.

3. Entropy generation analysis

For the flow of incompressible Jeffrey fluid under consideration, the volumetric entropy generation rate is given by,

$$E_v = \frac{K}{{T_0}^2}\left(\frac{dT}{dr}\right)^2 + \frac{\mu}{T_0}\left(\frac{dw}{dr}\right)^2 \tag{13}$$

The first term on RHS is on account of entropy generation due to heat transfer across a finite temperature difference, while the second term on RHS is for the entropy generation due to viscous effects.

The characteristic entropy generation number for the current case is given by,

$$E_c = \frac{K(T_1 - T_0)^2}{a^2 {T_0}^2} \tag{14}$$

The entropy generation number N_E in the form of dimensionless physical quantity is the ratio of volumetric entropy generation rate to the characteristic entropy generation rate [12,13].

$$N_E = \left(\frac{d\theta}{d\eta}\right)^2 + \frac{B_R}{T_P}\left(\frac{df}{d\eta}\right)^2 \tag{15}$$

where $T_P = \dfrac{T_1 - T_0}{T_0}$
Eq. (15) can be represented as

$$N_E = N_h + N_v \tag{16}$$

In Eq. (16), first term on RHS represents entropy generation due to heat transfer irreversibility and second term denotes entropy generation due to viscous dissipation.

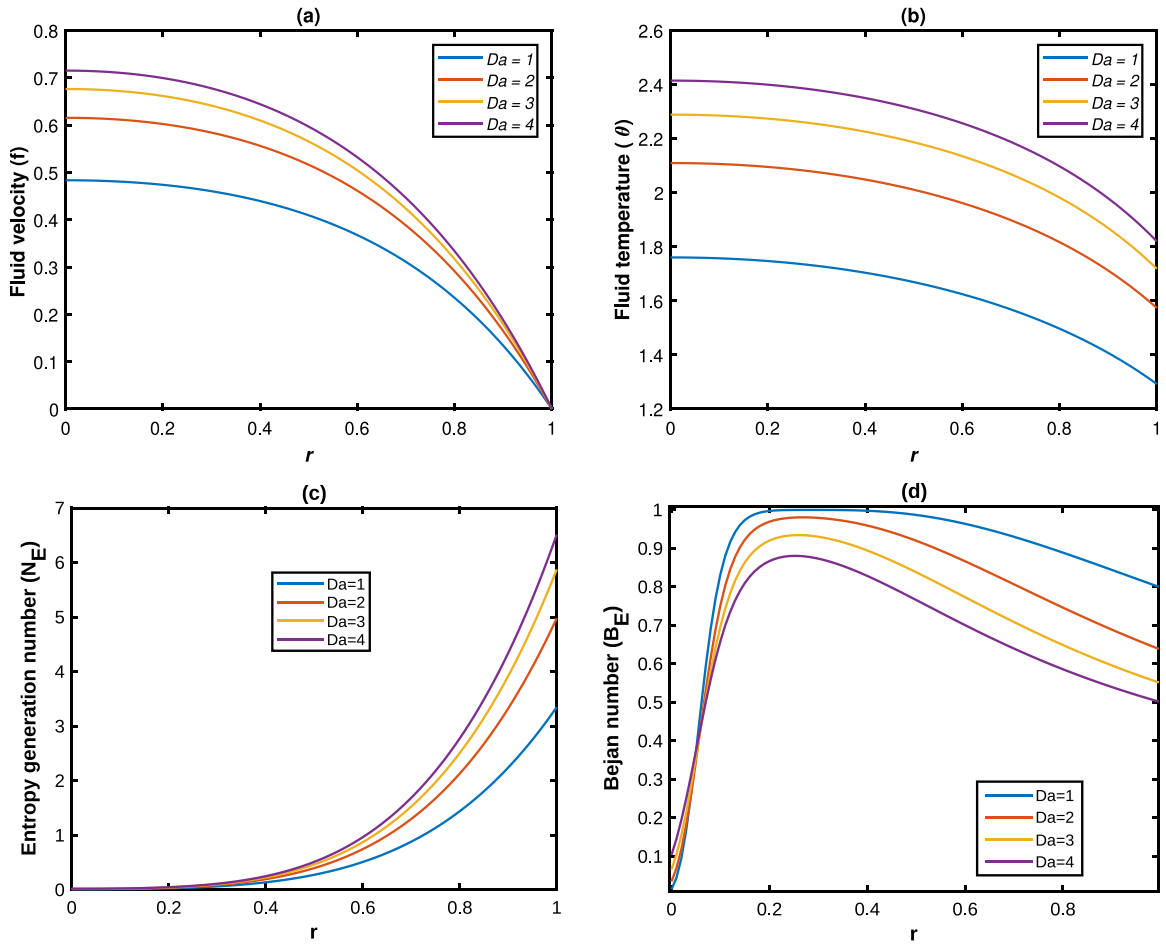


Fig. 6. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of Darcy number Da.

To determine irreversibility distribution, Bejan number (B_E), which is the ratio of entropy generation via heat transfer to the overall entropy generation, and is given by,

$$B_E = \frac{N_h}{N_h + N_v} \tag{17}$$

Bejan number always ranges from 0 to 1, where $B_E = 0$ denotes the dominance of irreversibility via viscous dissipation, $B_E = 1$ denotes dominance of heat transfer irreversibility, and $B_E = \frac{1}{2}$ indicates that the heat transfer irreversibility is equal to the fluid friction irreversibility in entropy generation.

4. Solution

The system of Eqs. (10)–(12) that governs the flow and heat transfer of the considered problem is of coupled and nonlinear nature. Due to the non-viability of the analytical solution, the finite difference method [50] has been implemented to acquire the numerical solution for the flow and heat transfer.

The detailed algorithm for the method of solution is presented in Appendix

The discretized version of the equations governing the flow (10) and heat transfer (11) along with associated conditions (12) are given as, for $i = 1, 2, \ldots, n-1$

$$G + [A - H_i]f_{i-1} - \left[2A + \left(M^2 + \frac{1}{Da}\right)\right] f_i + [A + H_i]f_{i+1} + C\theta_i = 0, \tag{18}$$

$$[D - E_i]\theta_{i-1} - 2D\theta_i + [D + E_i]\theta_{i+1} + F\left(f_{i+1} - f_{i-1}\right)^2 = 0 \tag{19}$$

along with the conditions,

$$\begin{aligned} f_i &= 0 \quad \text{at} \quad i = n, \\ \theta_i &= \frac{\frac{1}{h}}{\frac{1}{h} + Bi}\theta_{i-1} \quad \text{at} \quad i = n, \\ f_{i+1} &= f_i \quad \text{at} \quad i = 0, \\ \theta_{i+1} &= \theta_i \quad \text{at} \quad i = 0, \end{aligned} \tag{20}$$

where, $A = \frac{1}{(1+\lambda){h_1}^2}$, $H_i = \frac{\left(\frac{1}{(1+\lambda)\eta_i} - R\right)}{2h_1}$, $C = \frac{Gr}{Re}sin(\alpha)$, $D = \frac{1}{{h_1}^2}$, $E_i = \frac{\left(\frac{1}{\eta_i} - RPr\right)}{2h_1}$, $F = \frac{Br}{4{h_1}^2}$.

The system (18)–(20) is a nonlinear system of $(2n-2)$ algebraic equations in $(2n-2)$ unknowns. The Newton's method [51] has been implemented to find the numerical values of fluid velocities and fluid temperatures.

Once the numerical values of fluid velocities f and temperatures θ are obtained using aforementioned procedure, the entropy generation number and Bejan number can be subsequently found using (15) and (17).

5. Results and discussion

The flow and heat transfer results are provided for the MHD Jeffrey fluid flow considered in the present study. These results are obtained during the numerical simulations for different parameters that have been discerned in the upcoming discussion. The fluid flow velocity,

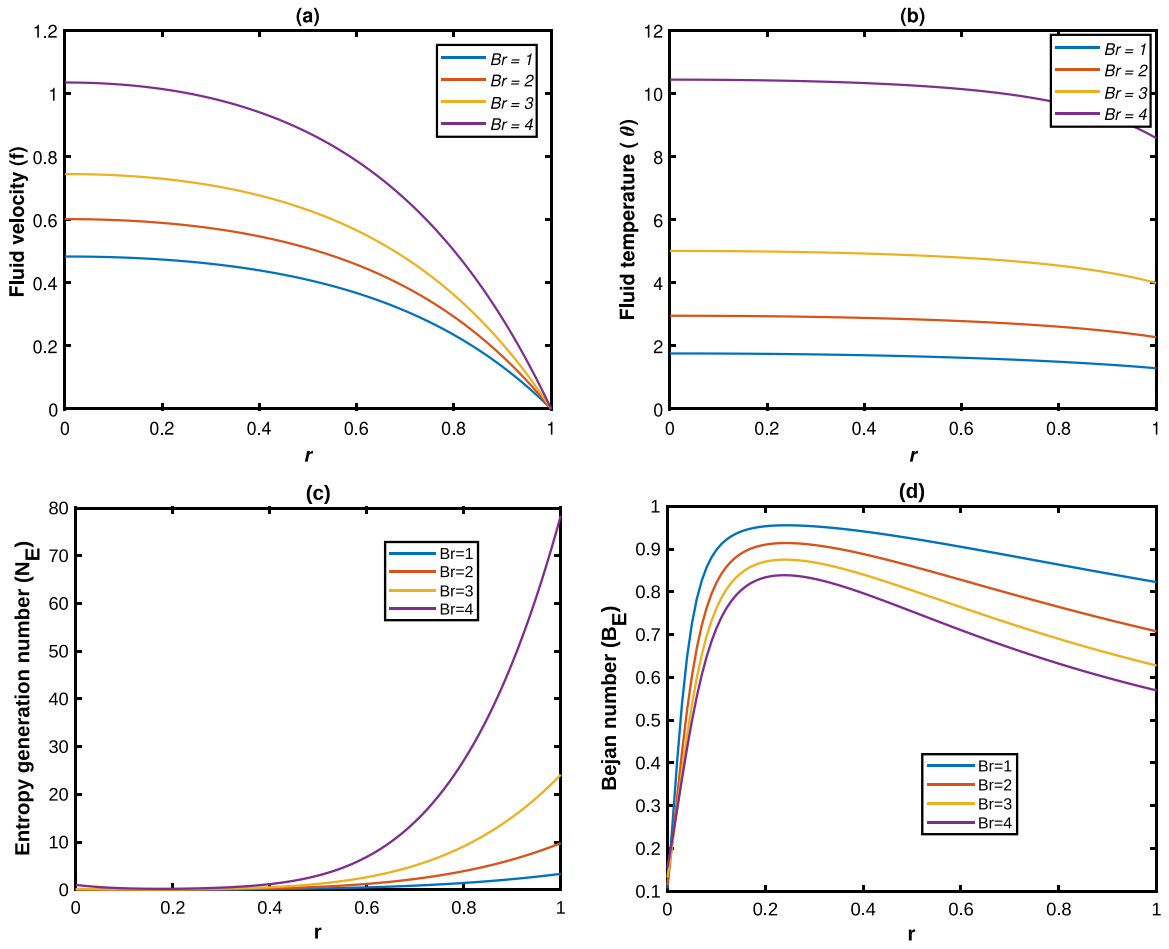


Fig. 7. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of Brinkman number Br.

temperature, entropy generation number, and Bejan number have been used to depict the variation in the various fields in the computational domain. The same has been shown for various values of channel inclination angle, Jeffrey fluid parameter, suction/injection parameter, Hartmann number, Darcy number, Brinkman number, Prandtl number, and Biot number. Furthermore, the pipe was assumed to be of infinite length. Accordingly, the results have been shown at a certain length where the fully developed condition was well achieved. While a certain parameter effect is being discussed, the others have been fixed at a certain value as: $\lambda = 1$, $\alpha = \pi/4$, $M = 1$, $Da = 1$, $Gr = 0.1$, $Re = 1$, $R = 1$, $Br = 1$, $Pr = 1$, and $Bi = 1$.

The authors want to declare that, the grid used in the present study has been put through the grid dependence test. It was seen that the present grid showed the same results as the next refined mesh (within an order of 10^{-3}). For brevity, these results have not been presented in the article.

In Fig. 2, variations of various flow parameters have been shown for different pipe inclination angles. It should be noted that $r = 0$ indicates the centre of the pipe, while $r = 1$ means that the value is being measured at the pipe surface. In subplot (a), it is clear that the velocity increases with the increase in pipe inclination angle, as the convection aids the motion of fluid inside the porous pipe. Further, as a consequence of this aided convection, a substantial rise in the temperature values is also witnessed in Fig. 2(b). The ongoing physics reflects differently on entropy generation, and Bejan numbers and different trends are seen. While N_E is zero at the centre and exponentially rises towards the walls, the Bejan number initially rises from the pipe's centre and gradually declines in value towards the walls.

The influence of various values of the Jeffrey fluid parameter is shown in Fig. 3. The effect of this parameter is more pronounced than the inclination angle. The velocity increases in the centre and gradually declines towards the walls, as seen in Fig. 3(a). The same trend is seen for different values of λ. With higher λ, it is noticed the centreline velocity augments and there is a positive rise in the kinetic energy. In the porous pipe, kinetic energy is used to overcome this centreline resistance; hence, at the walls, the velocity slowly reaches the value of unity. Furthermore, due to this increased kinetic energy with λ, the temperature is also seen to rise (Fig. 3(b)).

Again, the trends of both N_E and B_E match the previous trends, but they are more pronounced this time. Only at the walls does the N_E rise steeply. Also, B_E initially rises from the centre and then uniformly reduces. At the surface, they remain higher due to a large temperature gradient. It should be noted that the N_E shows a drastically significant value (about 25) at higher λ values.

In Fig. 4(a) and (b), when the suction on the hot wall of the porous pipe with simultaneous injection into the colder inner section of the pipe increases, both temperature and velocity increase. As discussed above, a consensual increase in the convection rate can be ascertained as the underlying physics behind this rise. Again, a qualitative similarity between the N_E and B_E trends is visible in subplots (c) and (d). However, in the case of B_E, the values for different R coincide at the wall at around the value of 0.45.

The influence of the Hartmann number is displayed in Fig. 5. When M increases, it is known that the resistive forces increase. Accordingly, a drop in the centre-line velocity value is seen with rising M, as depicted in Fig. 5(a). A similar trend is also seen in the temperature profiles due to the reduced convection rate. Qualitatively, the traits are the same or appear to be the same as before, but only the value declines instead of rising, as seen in Figs. 2–4.

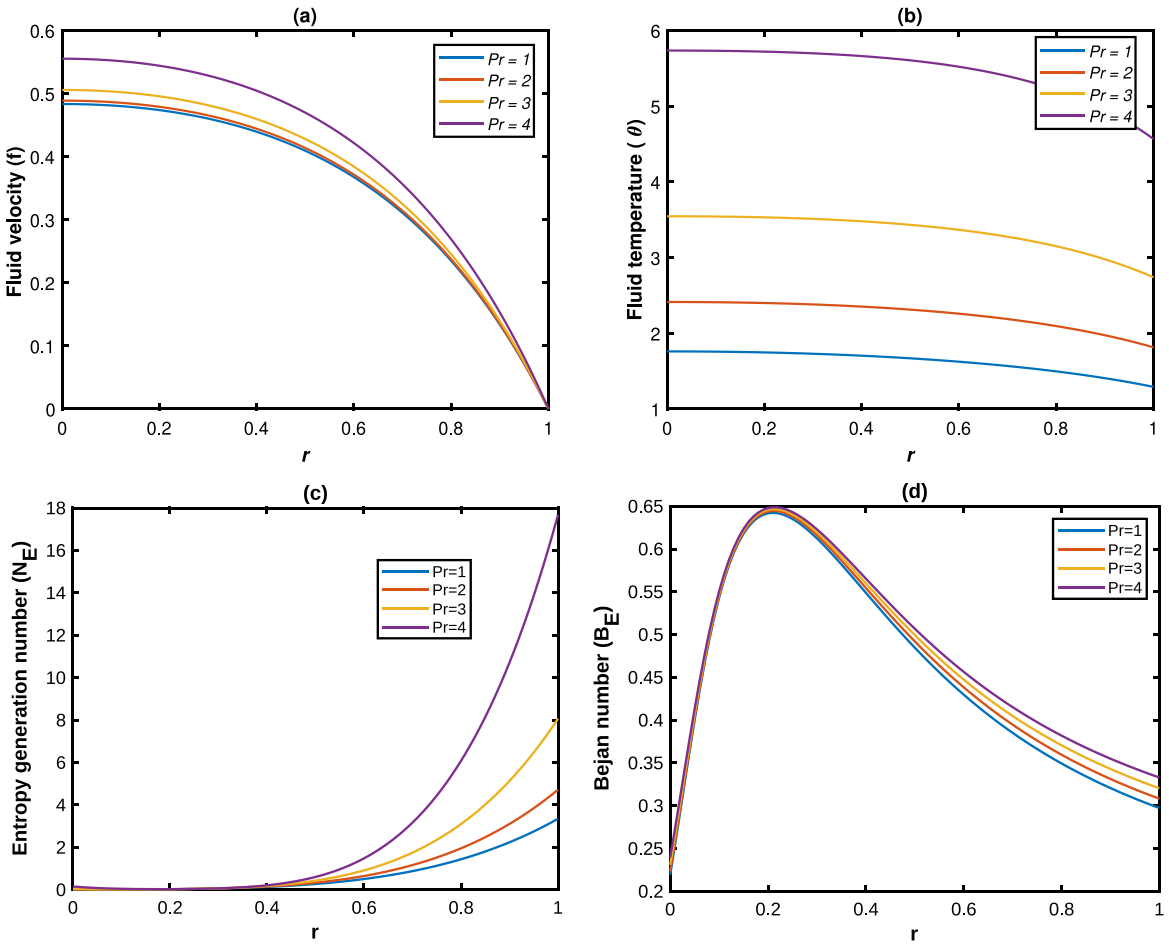


Fig. 8. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of Prandtl number Pr.

The entropy generation number decreases with the increasing M for velocity and temperature profiles, but the Bejan number shows the opposite trend (Fig. 5(c) and (d)), respectively. Perhaps the lower Hartmann number performs better in this case as it provided lesser overall resistance to the fluid flow, and consequently, the convection rate remains higher.

The effects of the Darcy number on the overall flow physics are shown in Fig. 6. Da implies the amount of viscous resistance that is offered to the fluid flow. The lower the Da, the higher the resistance, and vice versa. Fig. 6(a) agrees with this understanding, and the velocity increases on increasing the Darcy number. It is evident that when more fluid flows through the porous material, the heat dissipation rate increases, resulting in increased heat transfer. The temperature profiles tell the same story in Fig. 6(b). As the Darcy number attains a higher value, the heat exchange saturates, and the higher Da curves move closer to each other accordingly. Perhaps the overall porous media presence affects the overall irreversibility, and a decreasing trend in the Bejan number is witnessed in Fig. 6(d). N_E appears to be positively affected by Da, but the lines move closer at higher Da values, as discussed before. The values remain higher than the other parameters discussed before, except for λ.

Brinkman number influence on the ongoing phenomenon is shown in Fig. 7. Both the velocity and temperature profiles show a dramatic rise upon incrementing Br, especially the temperature profiles (seen in Fig. 7(a) and (b)). Viscous dissipation rises with Br, which results in enhancement in the overall temperature of the fluid. Therefore, a rise in buoyancy force is apparent, which causes the overall temperature trend to follow an exponential rise. However, entropy generation also increases with Br, especially near the surface of the pipe. This is because of the increased viscous effects and the consequential dominant friction at the wall. At the centre of the pipe, N_E is negligible compared to the surface of the pipe. Also, the effect is pronounced at higher Br values. The irreversibility, on the other hand, shows a different trait and decreases with Br, as displayed in Fig. 7(d).

Prandtl number effects are given in Fig. 8. It is a common observation that with increasing Pr, the overall velocity and temperature profiles show the declining trait. Here, with Jeffrey fluid, the opposite is reported to occur. An exponential growth with Pr in both velocity and temperature profiles is reported through Fig. 8(a) and (b). Prandtl number increment causes thinning of the thermal boundary layer and thickening of the hydrodynamic boundary layer. The same can be witnessed through the curvature of the profiles drawn in these subplots. The entropy generation number, however, increases rapidly with Pr, and B_E showed no drastic variation in Fig. 8(c) and (d). Biot number is known to be the ratio of the convection rate at the surface and the conduction rate within. Its trends in the present study are shown in Fig. 9. With an increment in Bi, an apparent reduction in the velocity is displayed in Fig. 9(a). This is because of the reduced heat transfer rate at the pipe walls. Again, a consequential convective cooling is witnessed through the temperature drop in Fig. 9(b). The entropy generation, as a result, is seen to reduce with Bi in Fig. (c). Nevertheless, a slight rise in the irreversibility is witnessed in Fig. 9(d) with B_E increasing.

6. Conclusion

The entropy generation analysis of the MHD steady flow of Jeffrey fluid through a pipe filled with porous medium has been studied in

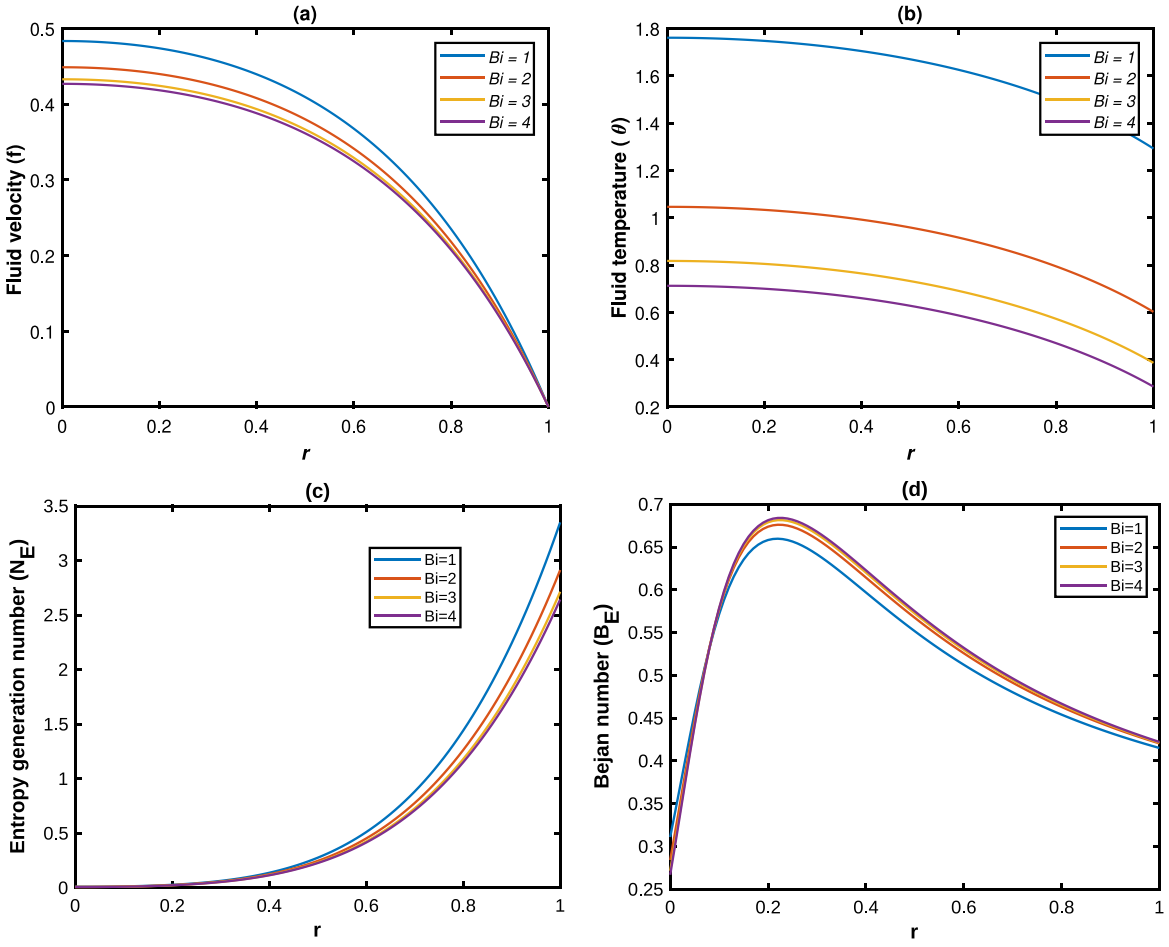


Fig. 9. Graphical results of (a) Fluid velocities, (b) Fluid temperatures, (c) Entropy generation number, and (d) Bejan number with varying values of Biot number Bi.

present research. The governing equations were solved along with the apt conditions. The major parameters of interest in this study are fluid velocity, temperature, entropy generation number and Bejan number.

The key findings of the present work are:

- The fluid velocity and temperature are found to be increasing with all the effects imposed in the study, except for magnetic effects and convective boundaries.
- With the application of magnetic effects and consideration of convective boundaries, flow and heat transfer tend to get suppressed.
- Horizontal inclination of pipe ($\alpha = 0$) hints at the reduced entropy generation as compared to vertical inclination ($\alpha = \pi/2$). Exact reverse trend has been perceived for irreversibility of the thermal system.
- Entropy generation is significant, especially at the boundary of the pipe, when Jeffrey parameter dominates.
- When there is neither suction nor injection, the irreversibilities due to fluid friction and due to heat transfer are almost equal. Naturally, suction suggests the reduced entropy and Bejan number and injection clues towards enhanced values in the same.
- Magnetic and porous effects impacts the entropy generation fields in exact opposite sense. Magnetic effects restrict and porous effects promote the same. Reverse trend has been noted in case of Bejan number.
- Viscous dissipation effects exponentially enhance the rate of entropy generation. Consequently the dominance of heat transfer irreversibility has been noted with greater viscous dissipation.
- Thermal resistances accounted through Biot number impact the entropy generation inversely. However, the magnitude of this change is insignificant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix

The solution methodology implemented in Section 4 is as follows:

Step 1: For the implementation of finite difference method, the domain [0,1] has been discretized uniformly with h_1 as the step-size in r-direction. The description of the discretized domain is $i = 0, 1, \ldots, n$, where $n = \frac{1}{h_1}$. Here $i = 0$ is the centre of the pipe and $i = n$ is the boundary of the pipe.

Step 2: Discretizing the system (10)–(12) by replacing the differential quotients with second order central difference approximations, viz.

$$X'' = \frac{X_{i+1} - 2X_i + X_{i-1}}{{h_1}^2}$$

and

$$X' = \frac{X_{i+1} - X_{i-1}}{2h_1}$$

Step 3: The resultant system (18)–(20) is the system of $(2n-2)$ nonlinear algebraic equations in $(2n-2)$ unknowns of the form

$$F(x) = 0.$$

Step 4: Taking X_0 as initial guess, this nonlinear system of equations has been solved using Newton's method for system of nonlinear equations for $k = 0, 1, 2, \ldots \ldots,$

$$X_{k+1} = X_k - [F'(x_k)]^{-1} F(x_k).$$

This iteration scheme has been implemented till the solution of desired accuracy is obtained.

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